

Institute of Actuaries of India

Subject SA4 – Pensions and Other Benefits

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) a) The ultimate cost of the plan is not known until the last benefits are paid out. Until then, estimates of the obligation are made at various points in time. Estimates depend on the actuarial method and assumptions chosen. The different approaches and assumptions will determine the pace and timing of contributions and not the ultimate cost itself. [1]
- Different actuarial methods can be used depending on the purpose of the valuation. E.g. Accounting valuations; Discontinuance valuations or funding valuations [0.5]
 - Accounting valuations are for financial reporting only and largely dictated by the accounting standards in a country e.g. IndAS19 / AS15 (Revised). The purpose is to estimate the obligation and resulting P&L cost on a fair value best estimate basis, assuming a going concern principle. Most typically, the discount rate is market related and valuation method is prescribed. Often this type of valuation may also be used in a corporate transaction assessment. [0.5]
 - A Discontinuance valuation's purpose is to assess the solvency of the plan assuming all benefits accrued at the valuation date needed to be paid/secured immediately. This is useful in a wind up scenario and may not be applicable in the given scenario. [0.5]
 - A funding valuation helps the sponsor / Trustees determine appropriate contribution requirements and consider the long term sustainability of the plan. There will be greater flexibility in determining appropriate assumptions compared to an accounting valuation. For the given scenario, this will be the base approach that will guide the setting of assumptions and actuarial method. [0.5]
- [Max 3]**
- b) India does not have a formal legal requirement for funding valuations in pension plans. It means there are no specific restrictions as far as:
- minimum level of contributions
 - specified assumptions to be used
 - pace of funding any shortfalls or amortising surpluses
 - actuarial method to be used [1]
 - However, the actuary will need to adhere to Actuarial Practise Standard 27 in developing the advice related to the funding valuation. It means that there will be guiding principles the actuary will use to develop the assumptions and actuarial method. [1]
 - There are Income Tax rules that need to be considered when setting the contribution rates. Per the Income Tax Act, the total contributions that can be paid to a Provident Fund and Superannuation plan put together is a maximum of 27% of total wages. With the statutory PF at 12%, this means a maximum of 15% to the superannuation plan. However, it is not clear if this would apply to DC schemes only, or DB schemes as well. Further the restriction is for tax benefit purposes. [1]
- [Max 3]**
- c) Overall principles for assumption setting needs to:
- reflect the level of prudence the stakeholders wish to have [0.25]
 - financial assumptions should broadly demonstrate a sense of consistency between them [0.25]

- **Demographic Assumptions**

Mortality pre-retirement: Given the scheme will not be large enough for credible analysis of its own mortality experience. It would be recommended to use a latest standard table. [0.25]

Mortality post retirement: As the plan pays its own pensions, this assumption is more significant. One could use a standard table, ideally one that reflects the experience of retirees. E.g. LIC Annuitants 1996-1998 table [0.25]

Mortality improvements: As the duration of the plan is long and retiree payments are made from the plan itself, mortality improvement assumption should be incorporated. This can be done by adjusting the current tables by an age rating or percentage. Alternatively, one can apply a year on year improvement factor such as 1% p.a. improvement on each probability qx. [0.25]

Attrition: With the 10 year vesting on withdrawal benefits and immediate pension this is can be a significant assumption. Normally one may consider the past experience of the company and also industry experience. [0.25]

However, one must also consider that the scheme is now closing and so the experience of the continuing members needs to be reflected. It may mean that there may now be lower attrition amongst the members as compared to the overall company, especially those with less than 10 years' service. [0.25]

Early retirement: Extending the attrition assumption – as there are benefits immediately, and no reduction factor post age 60, one should consider the attrition/early retirement assumption for post 55 years of age extra carefully. [0.25]

Other: other assumptions such as percentage married / spouse age can be kept from typical market practice [0.25]

Other: Commutation take up rate; as the conversion rate is fixed, the significance of the assumption depends on the comparative value per the valuation basis of a pension [0.50]

- **Financial Assumptions**

Discount rate (i): Need to consider the consistency of valuation of assets and liability. [0.25]

One can use a market related approach or a long term value approach [0.25]

The discount rate should reflect the weighted expected returns from the current and/or the future benchmark investment asset allocation strategy. This is especially true, given the plan is self-managed. [0.25]

Due to Indian regulations, the asset allocation will be mainly fixed income (Central Government and State Government/PSU Bonds) with some equity exposure, up to the maximum allowable of 15% of contributions. [0.25]

Salary Growth (e): Reflect the general economic scenario, as well as the experience and plans of the company itself. May consider consistency with discount rate [0.25]

The assumption needs to reflect general inflationary increases and promotional increases [0.25]
[Max 3]

d) One would recommend an Attained Age method (AAM) for the following reasons: [0.25]

- We are conducting the valuation as at the time of closing the plan and there will be no new entrants [0.25]
- AAM provides for a reasonable level of security, given that future salary growth is included in the past service liability. [0.5]
- The AAM standard contribution rate formula includes potential future service and calculated at the time of closure, technically the SCR does not need to be recalculated at each subsequent valuation. [1]
- Future valuations, will then focus on the amortisation of surplus/deficits. If assumptions are borne out in practice then a AAM modified contribution rate will result in a more stable contribution pace of funding [0.5]
- Initially, the AAM will require higher contributions than a PUC method (assuming typical assumptions where $i > e$). However, this balances off with AAM building up surpluses that can be amortised for a more stable overall modified rate. [1]
[Max 3]

e) Other considerations

- Level of prudence may need to be higher, given the balance required of a longer term 90% target with a quarterly check point of not dropping below 80%. [1]
- This may also lend itself to adopting an alternative approach than market related assumptions and asset valuation method. [0.25]
- The degree of prudence needs to be decided between the Sponsor and Trustees. [0.25]
- This may depend on the level of Sponsor covenant. If the Sponsor is strong and committed to the plan then one may have a lower level of prudence in the assumptions [0.5]
- One will need to take into consideration the level of cash flow the Sponsor is willing to make realistically. [0.25]
- The valuation should be done on a range of assumptions [0.25]
- If there is a higher level of equity exposure then one may include a higher degree of prudence [0.25]
- There is no time period mentioned for the 90% target – this will need to be decided and how frequently that will get measured [0.25]
[Max 3]

ii) Investigations

With current valuation

- Start with the current funding valuation [0.25]
- Ensure a number of sensitivity scenarios are done on the discount rate for different model benchmark portfolios [0.5]
- Looking at the contribution requirements should assets fall and rise by a certain level as one offs to test the durability of funding levels. This will be important in a market value valuation approach and given the strict funding level governance the sponsor is applying quarterly. [0.5]

Further Asset-Liability study

- Extend the current valuation and plot out the expected cash-flows from the assets and the liability to test the extent of duration mismatch. [0.5]
- From this, one needs to decide if the objectives are duration matching; cash-flow matching or the balance of risk vs returns [0.5]
- As the scheme is closed, it will mature quickly and so reinvestment risk can be measured by projecting future valuations and contribution requirements under different assumption scenarios [0.5]
- Ideally this should be done on a stochastic approach; however; cost and practicality may dictate a deterministic approach is used. [0.5]
- As the scheme will mature, the cash flows projection will indicate future liquidity requirements. This will also impact the investment strategy. [0.5]

Specific ALM points for assumptions

- One may start with more a best estimate basis as a central assumption set [0.25]
- For the cash flow projections, a discount rate will not play a significant role [0.5]
- However, for projected valuations one may choose discount rates that are term dependent especially with shortening durations of the closed scheme in the future [0.5]
- The demographic assumption impacts the timing of cash flows and that is important for an ALM study. [0.5]
- It means assumptions like the early retirement become more significant. [0.25]
- The percentage of commuted value taken will play a larger role than in the funding valuation [0.25]
- For the investments, the view on future interest rates, will be important due to reinvestment of proceeds. Future potential changes in investment asset allocation benchmarks for the reinvestment rates will play a more significant role in an ALM [1]

[Max 7]

- iii) Schemes may choose to self-manage their investments or have an insurer managed product. [0.5]

Self-Managed pattern

Effective from 1 April, 2015:-

Category	Percentage
Government Securities and Related Investments a) Government Securities, b) Other securities defined where the principal whereof and interest whereon is fully and unconditionally guaranteed by the Central Government or any State Government. (s.t. 10% maximum of total fund) c) Units of mutual funds set up dedicated for investments in Govt securities (s.t. 5% maximum of total fund/ accretions)	Minimum 45% and up to 50%
Debt Instruments and Related Investments a) Listed (or proposed to be listed in case of fresh issue) debt securities issued by bodies corporate, including banks and public financial institutions, which have a minimum residual maturity period of three years from the date of investment. b) Certain defined commercial Bank debt; Rupee bonds; Term deposits	Minimum 35% and up to 45%

c) Units of Debt Mutual Funds as regulated by Securities and Exchange Board of India (s.t. 5% maximum of total fund/accretions) d) Some Infrastructure bonds	
Short-term Debt Instruments and Related Investments a) Money market instruments b) Term deposits c) Liquid mutual funds Restrictions apply to quality and term of some of these categories	Up to 5%
Equities and Related Investments a) Shares of body corporates listed on Bombay Stock Exchange (BSE) or National Stock Exchange (NSE), which have: (i) Market capitalization of not less than Rs. 5000 crore as on the date of investment; and (ii) Derivatives with the shares as underlying, traded in either of the two stock exchanges. b) Units of mutual funds regulated by the Securities and Exchange Board of India, which have minimum 65% of their investment in shares of body corporates listed on BSE or NSE. (s.t. 5% maximum of total fund / accretions) c) Certain index tracking ETFs d) ETFs constructed specifically for disinvestment of shareholding of the Government of India in body corporates e) Certain Derivatives, only for hedging (s.t. 5% maximum of total fund in sub category b) to e)	Minimum 5% and up to 15%
Asset Backed, Trust Structured and Miscellaneous Investments a) Commercial mortgage based securities b) Units of REITs c) Units of Infrastructure investment trusts d) Asset based securities regulated by SBI Restrictions apply on quality assess by rating agencies	Up to 5%

[up to 1 for category 0.5 for percentage]

Trading allowable within the Turnover ratio of two.

[0.25]

No more than 5% of accretions in any year should be made in self investment (ie. Company where the sponsor holds more than 10% of that invested company)

[0.25]

Insurer managed

In addition to this the scheme has the option of having some or all the portion of the assets managed by an insurance company. The investment compliance is then governed by the insurer regulations it has itself with the IRDAI. In practice this means more flexibility in investment strategy and exposure.

- Choice of traditional deposit administration products whereby there are some minimum rates of interest guaranteed, the capital and accumulated corpus value is not at risk. Only future declared interest rates may change. [0.5]
- Choice of unitised funds managed by the insurer. They would have a range of asset allocation based funds e.g. Growth; Balanced; Debt etc.. [0.5]

One would need to check the current Trust Deed and Rules to see if there are any restrictions of how the funds may be invested. If any changes are required to the Deed, a Deed of Variation can be filed to reflect the same. [0.5]

[Max 9]

- iv) The pattern for a PF Trust is largely similar [0.25]

The pattern of PF is said to be governed by notification issued by Ministry of Labour [0.25]

Key Differences are:

- Debt Instruments and Related Investments minimum is 20% not 35% [0.5]

[Max 1]**v) Commutation**

Up to one-third of the pension can be commuted and also the commuted value is restored at age 75

Impact on the funding status will depend on:

- Funding assumptions and type of valuation one is considering [0.25]
- How the Rs25 value for each Rs1 annual pension compares to the actuarially assessed value of the pension given up. [0.5]
- Under the current accounting valuation the reserve for a retiree at the point of age 60 is:

- a. Pension x annuity value at age 60 (using scheme assumptions)
In reality at age 60 a member is benefiting from [1]

- b. $\frac{1}{3} \times \text{pension} \times (25 + \text{deferred annuity value at age 75})$
+ $\frac{2}{3} \times \text{pension} \times \text{annuity value at age 60}$

If b) > a) then funding status under the accounting basis reduces and actuarial losses are incurred.

Early Retirement

The scheme pays immediate pension if completed 10 years' service. There is a reduction factor of 3% per annum if the pension starts earlier than age 55.

With early retirement, key aspects are a factor:

- a. The pension is paid earlier so less time for assets to build [0.25]
- b. The pension is paid for longer [0.25]
- c. The pension is less due to the salary increments on past service not having for as long [0.25]

The impact of 60% retiring on average at age 55 on the funding status will depend on:

- The assumption relation of the discount rate and salary growth [0.5]
- The closer the two assumptions are the lesser the impact [0.25]

- If salary growth higher than discount rate then an earlier start actually will improve the funding status and vice versa [0.5]
- The current adjustment factor compensates somewhat for the pension to be paid for longer in terms of mortality. However, it is not applied for the first 10 years before NRA. It means that this would lower the funding status. [0.5]

[Max 4]

vi) General

One should first check how the factors set out in the Trust Deed and Rules (TD&R) have been arrived at. Further how the factors to be decided are set out in the TD&R. This may indicate to what extent actuarial advice is required, or the base principles they need to adhere to, or not say anything. [1]

If any rates are set in the TD&R then these will need to be updated before implementing any new rates. [0.25]

It would be recommended to have the clauses to be flexible to allow regular reviews and update the factors as necessary, upon actuarial advice. [0.25]

The TD&R should also be worded in a way that the options can only be exercised upon “consent” of the Sponsor and/or Trustees. This also gives greater control on scheme should the need arise. [0.5]

Commutation Factor

Starting point should be an actuarial value calculation of the pension being given up. [0.5]

Key scheme features that need to decide if giving up or not.

- a. The restoration portion: Given the scheme rules, this should not be in the commutation factor. It means the factor should be for a value up to age 75 only. [1]
- b. Spouse’s pension: Administratively it would be easier if the spouse’s pension was included in the commutation factor and therefore the spouse’s pension is then based on the pension actually being paid at the time of death. [1]
- c. There is no mention of discretionary benefits but one should check and decide if these are to be incorporated; or be very clear that they are not e.g. ad-hoc cost of living adjustments. [0.5]

Actuarial value consideration

- a) On what discount rate will the equivalence be done? [0.25]
This depends on which valuation basis the Sponsor/Trustees want to equate. A funding basis may enable a longer term approach and also incorporate some prudence. If this is higher typically than an accounting basis then the commuted value will be lower for members, thereby being less attractive. However, it makes it more future proof and more equitable between generation of members – especially in a closed scheme. [1]

A market related discount rate will reflect the “cash” cost better at any point in time and possibly be more relatable to the members. [0.5]

Another approach may be to reflect a rate that members may be able to generate as returns from the commuted value themselves. E.g. senior citizen deposit rates [0.5]

A slightly higher discount rate may be used to reflect the perceived benefit the immediate cash has compared to the longer term value. [0.5]

- b) The rates should be age related to reflect the differing length of time of pension being given up. This will need to be balanced with administrative ease. [0.5]
- c) Mortality tables should be those applicable to post retirement. [0.25]
- d) As we are multiplying the factor to the pension to get the lump sum value; one may propose a prudence measure and not include mortality improvements into the factor. [0.5]

Early Retirement

At first glance it may seem the early retirement benefits are generous for members. [0.5]

- a) The reduction rate is 3% p.a. applying to pre age 55 only. [0.25]
- b) However, this does not explicitly consider the lost interest and reduced future salary growth. If the salary growth assumption is higher than the discount rate (which in India is not unusual) then the factor should be lower (e.g. if salary is 2% higher than discount rate) then each year someone retires early saves a net 2%. For actuarial equivalence the factor should actually increase the early pension. [1]

However, a prudence may be applied here and the pension not actually increased. [0.5]

This is compensated for members currently by not reducing the pension at all from age 55 to 65. [0.5]

- c) One could implement a compound factor approach rather than simple. This may help the scheme in long term as immediate pensions are paid irrespective of age at leaving (as long as 10 years' service completed). Much younger retirees then will get far lower pension using a simple rate as a simple rate is more penal. Currently therefore, the scheme actually benefits. [0.5]
- d) The early retirement factors should be reviewed if the funding actuarial basis is significantly updated and the i-e gap changes significantly. Also if there are significant changes in the mortality tables. [0.5]
- e) Keeping an administratively simple calculation is better than strict actuarial equivalence for both the members and the pension scheme administrators. [0.25]
- f) If the Sponsor wanted to control the level of early retirements for managing manpower numbers, then one should introduce an employer consent requirement for early retirement. [0.5]
- g) All things considered assuming i-e gap of about 2% ($e < i$), one could argue that the existing factor is actually not too far off and need not change. [0.5]

This is of course very dependent on i and e assumptions. [0.5]

- h) However, given the current level of funding and the popularity of the early retirement one may consider tweaking the factor slightly to help the funding position e.g. say 1.5% per annum reduction from age 65 to 55. [0.5]

[Max 14]
[50 Marks]

Solution 2:

- i)
- A final salary pension scheme is exposed to same risks whether or not it is closed to new entrants, i.e.
 - Salary escalation is more than expected;
 - Price inflation is more than expected;
 - Investment return is less than expected;
 - Pre and post retirement mortality is less than expected;

- Family size of members is more than expected [0.25 for each point]	[Max 1]
• Such a scheme when closed to new entrants has additional risks due to its changing profile as it has lesser time now to manage such risks	[0.5]
The changing profile & the risks associated with such a scheme are as under:	
1. Changing membership profile:	[0.25]
- At present, the proportion of pensioners in the Scheme is 52%. This proportion is expected to increase in future as more members move to the Pensioner Category without corresponding increase in active members.	[0.5]
- The proportion of active members (48%) will decline rapidly over time as they approach the NRA.	[0.25]
- The average age of the members of the scheme will continue to move upwards	[0.25]
- At some point all members of scheme may be pensioners only	[0.25]
- Membership may finally decline to a few members	[0.25]
2. Changing benefit/liability profile:	[0.25]
- The past service benefit of active members which is currently around 69% will increase initially as these members add additional service & come closure to their NRA but shall decline thereafter as these members move to pensioners category ultimately	[0.5]
- Pension increases of pensioners are linked to price inflation whereas pension benefit of active members is linked to salary inflation due to final salary link. Though liabilities of the scheme may remain real, the salary inflation link will move to price inflation over time	[1]
- The changing liability profile of the scheme poses additional challenges in investment management of the scheme	[0.5]
- The liability profile as it matures will need more monitoring and action by employer/Trustees	[0.25]
3. Changing contribution needs:	[0.25]
- The changing membership profile may result in increase in the rate of contribution as % of salary of active members	[0.5]
- However, the contribution in monetary terms may reduce over time	[0.25]
4. Changing funding needs:	[0.25]
- So long the active members are sufficient in number, the scheme will have some flexibility in its funding needs	[0.5]
- However, when proportion of active members reduce, such flexibility reduces and when all members move to pensioner's category, scheme needs to be sufficiently funded	[0.5]
- Otherwise the scheme will move from funded (or partly funded) to pay-as-you-go scheme	[0.25]
5. Changing valuation requirements:	[0.25]
- Initially the employer may need valuation for both the funding and accounting	[0.25]

	purposes	
	- However, as all active members move to pensioner's category, there may not be any service cost in valuation for accounting purposes	[0.25]
	- Over time the interest cost component will form the majority of the pension cost, under valuation for accounting purposes especially when the scheme continues to be partly funded	[0.5]
	6. Changing valuation requirements:	[0.25]
	- Presently large part of liability is linked to salary inflation which will ultimately may move to price inflation. The Trustees need to take this into account while making investment decisions	[0.5]
	- As the scheme matures, liquidity and security aspects will gain more importance in investment decisions	[0.25]
	- The asset-liability mismatch, if any needs to be monitored more closely as the scheme matures in future	[0.25]
	7. Scheme administration:	[0.25]
	- Initially the scheme administration will continue without much change but as the scheme matures, the work will reduce and time will come when there will be few pensioners whose pension needs to be paid	[0.5]
	- When there are few pensioners, Trustees may consider to purchase annuities from an insurer	[0.25]
	- However, as the pensions are linked to price inflation and it is unlikely that the insurers will provide inflation linked annuities, the Trustees will need to purchase additional pensions periodically for increase in inflation	[0.5]
	8. Increasing expenses:	[0.25]
	- As the scheme matures, more monitoring will need to be done involving higher expenses	[0.25]
	However, when the scheme matures expenses will start falling	[0.25]
		[Max 12]
ii)	Factors to be considered & the Approach:	
	• By making such an offer, the objective of the PSO might be to	
	- Reduce the uncertainty due to mortality (longevity), interest rates and price inflation (or rate of pension increases);	[0.5]
	- Save some of the costs; and	[0.25]
	- Reduce the immediate outgo of funds from the scheme	[0.25]
	• The offer is similar to commutation of pension where payment of lump sum is deferred till death	[0.5]
	• It will need calculation of factors to be multiplied with the amount of pension forgone to give the lump sum to be paid on death of the pensioner.	[0.5]
	• To calculate these factors, the PSO needs to decide	
	- Whether or not the factors should be age based;	[0.25]
	- Whether or not the factors need to be based on gender;	[0.25]
	- Whether or not the pension increases need to be considered	[0.25]
	• If factors are not age based, then it gives scope for anti-selection (lower aged pensioners not opting).	[0.5]
	• Hence it would be better to calculate age based factors	[0.25]

- If gender based factors are to be derived, then separate factors need to be calculated for males and females based on different mortality assumptions [0.25]
- It will increase administrative complexity [0.5]
- Also consideration of pension increases will reduce uncertainty due to future inflation [0.25]

The factors (F) may then be calculated as $F = \left(\frac{a_x^{(12)}}{A'_x} \right)$, where

- x is the age of the pensioner at the time of option,
- a_x is the annuity factor calculated at (i-p) rate, where i is discount rate & p is the rate of pension increases considered; and
- A'_x is the Assurance factor calculated at i discount rate [2]
- The above factors consider pension increases. If pension increases are not to be considered, then "ax" is to be calculated at i rate of discount [0.5]
- The PSO has further to decide the discount rate (i), rate of pension increases (p) and the post retirement mortality rate to be considered for calculation of factors [0.5]
- The mortality assumption for the time to death will be considered for an average pensioner whereas pensioners will exercise option considering their own personal circumstances, for example [0.5]
- If pensioner's health is critical or bad, then he will be inclined to opt for it [0.25]
- If his health is good and he expects to survive for a long period, he may not opt in favour of lump sum unless the same is attractive [0.5]
- Also the single pensioners without dependents may not opt as they would like to enjoy their full pension themselves [0.5]
- Alternatively PSO may consider health of the pensioner and apply mortality adjustment based on medical underwriting of pensioners exercising option [0.5]
- However, medical underwriting may complicate the option process and may be found full of hassles and not feasible [0.25]
- If the terms offered are attractive, then the take up rate may be high [0.25]
- But then the objective of the offer may get defeated as PSO may not benefit from the offer [0.25]
- The basis therefore needs to be best estimate and fair so that neither the PSO loses nor the pensioners and the offer becomes successful [0.25]
- So the discount rate needs to be based on the long term return expected on pension fund investments; [0.25]
- Pension increases assumption should reflect the expected movement in long term price inflation [0.25]

Other factors to be considered by the PSO are:

- Whether or not family pension should be considered
- Whether the offer needs to be limited to member pensioners or should include family pensioners as well [0.25]
- If family pension is considered, then different factors need to be worked out [0.25]
- One applicable to single pensioners, and [0.25]
- The other applicable to pensioners with spouse alive at the time of option [0.25]
- Alternatively factors may be worked out considering proportion of pensioners having spouse alive at option time and offer a single factor for all [0.25]
- If single factors are used, then single pensioners may get higher benefit [0.25]
- As single pensioners are not likely going to opt for it (as they may like to enjoy full benefit during their life time) unless attractive to be left for children, PSO may consider to offer a single factor for both the single & non-single pensioners [0.5]

	The option should be available only for a limited period (say, 3 months) to serve the objectives.	[0.5]
	PSO will need to discuss with Trustees if the Scheme is established under Trust & amend the scheme Rules.	[0.5]
		[Max 15]
iii)	Being a pension plan the PSO needs to consider the current retirees and also the future retirees	[0.25]
	Risks and Administrative matters:	
	Buying annuity from an insurance company removes the risks such as longevity, investment from the fund and transfers them to the insurance company.	[0.5]
	This may help to control the deficit through the release of prudent margins, if any built in the funding valuation	[0.5]
	But such transfer of risk through annuity purchases will come at a price as the insurance company will be charging for the cost of the risks being transferred and for the expenses & profit margins.	[0.5]
	Further, the members are entitled for pension linked to price index and it is unlikely for an insurer in India to provide such annuities. The Trustees therefore need to calculate changes in annuities every time there is an increase in annuities due to movement of price index and purchase additional annuities	[0.5]
	Further the new purchases for additional annuities will be at the price prevailing at the time of purchase. It means the risks are not transferred fully to the insurer.	[0.25]
	This will therefore make the process complicated	[0.25]
	Further Complications may arise when the price index comes down in which case the annuities will need to be surrendered	[0.5]
	The PSO may need to revisit the valuation, funding, financing and investment strategy of the Scheme.	[0.5]
	PSO will still be responsible for the administration, overall governance and accountability of the scheme	[0.25]
	Consider the disadvantages of such purchases also	
	- Removes the benefit of any favourable experiences (e.g. investment)	[0.25]
	- Variability of the annuity purchase price and the taxes on such purchases	[0.25]
	- Non availability of the annuities to match the promised pensions such as restoration of commuted pension after some age or annuities at younger ages	[0.25]
	- Any default by the insurance companies may lead to complaint, legal/regulatory issues for the PSO	[0.25]
	Any change will need be communicated to the retirees	[0.25]
	Retirees will come to the PSO for queries and support	[0.25]
	The Trustees will need to internally keep track of the restoration and further purchase of annuities	[0.25]
	Investment related matters:	
	Immediate liquidity needs will increase drastically	[0.25]
	It is unlikely that liquidity will exist to pay the retirees buyout immediately	[0.25]
	This will mean either significant sell off or a substantial shortfall in contributions	[0.25]
	Future duration and liquidity needs should need to be factored in to investment strategy	[0.25]
	An asset-liability study will be crucial for these reasons	[0.25]
	Financial matters	
	Purchase of annuities from insurer does not eliminate the liability from the Trustees	[0.25]

- An assumption of future annuity rates will need to be considered in future valuations [0.25]
 - Any difference in the immediate cost of annuities for retirees compared to accounting valuation reserve will lead to an immediate settlement cost or credit that needs to be recognised in the PSO's P&L [0.5]
 - For future retirees any difference would tend to flow as experience item under AS15 (Revised) in P&L but in IndAS19 it will flow in to the OCI statement [0.5]
 - There will be change in the duration due to which change in discount rates in the accounting valuations [0.5]
- [Max 8]**
- iv)
1. Increase the NRA from 60 to 62 or 65 as the maximum pension earned by a member is 50% of final salary. It will defer the payment of pension for few years [0.5]
 - However, those who have earned less than 50%, will add further [0.25]
 - Also the pension may increase due to increase in salary at extended NRA [0.25]
 2. Changes in the benefit design (in respect of future service)
 - Lower accrual rate, say instead of 1/66 to 1%;
 - Modify definition of pensionable salary (e.g. last 5 or 10 years' salary average or career average);
 - Ceiling on total number of years' service that counts for pension;
 - Ceiling on the maximum pension;
 - Ceiling on Pension increases;
 - Revisit the components of pensionable salary (e.g. some allowances, if any can be removed)
 - Redefine the definition of "family" for dependent pension [0.25 for each point] [Max 2]
 3. Enhance the vesting age or vesting service conditions, if any [0.25]
 4. Close the scheme for future accrual of benefits or at least encourage active members to move to DC scheme in respect of future service [0.5]
 5. Revisit early and/or late retirement terms, if any [0.25]
 6. Encourage early retirement of active members – it removes future accrual but commences pension at earlier date than NRA [0.5]
 7. Revalidate the assumptions used in valuation to remove the "excess" prudence built in valuation, if any [0.5]
 8. Revisit commutation factors [0.25]
 9. A combination of the above [0.25]
- [Max 5]**
- v) Why a funding contribution rate should not be used as cost for CTC purpose:
- Gratuity is a defined, Statutory, long term benefit plan, financed fully by the Employer. The recommended Contribution rate (of 17% of AWB) reflects the contribution requirement for the year 2018-19, to meet the funding objectives of the plan. [0.5]
 - The recommended rate represents the weighted average of the one year estimated funding contribution rate of the individual members [0.5]

- It will be more or less than the one year cost of accrual of the individual members [0.5]
- So the scheme's recommended rate should not be used as a cost of individual employees for their CTC purpose [0.5]
- Even if funding rate is calculated for individual employees, the same should not be used as cost for CTC due to following reasons:
 - The recommended contribution rate reflects the funding objective of the plan. For e.g. if the PSO is having a funding target (of less than 100%, say 80%), then using the same for individual CTC purpose may understate the one year accrual cost of the members [1]
 - There may be surplus or deficit in the fund and the recommended rate would have considered the same. It would thus not be appropriate to use it for CTC purpose [0.5]
 - The salary growth assumption may over-state or understate the salary growth of the individual members. [0.5]
 - The decrements used in valuation such as withdrawal, death are not appropriate to the surviving individual members. [0.5]
 - Also the funding advice may use margins in the assumptions and in that case the recommended rate may overstate the cost [0.5]
- Hence whenever there is a request from the employer to give contribution rate for individual employees, the actuary should ask the reason thereof and if the same is required to be used as cost for CTC purpose, then the actuary should advise for separate valuation of cost for CTC [1]

Contribution rate for CTC purpose:

- A contribution rate for an employee for his CTC purpose is the cost of the benefit accrued during that period by that particular employee [0.5]
- The cost of each year's accrual of benefit may be calculated as Standard Contribution Rate (SCR) using Projected Unit Method (PUM) – PUMSCR [0.5]
- However, where the discount rate (i) is greater than salary escalation (e), the PUMSCR will increase with age of employee and vice versa [0.5]
- Therefore PUMSCR will need to be calculated for the employees every year. If the employer wants to have a stable cost, then PUMSCR may not work [0.5]
- A stable cost for existing employees may be calculated as SCR using Attained Age Method (AAM) i.e. AAMSCR [0.5]
- Similarly, a stable cost for new employees may be calculated as SCR using Entry Age Method (EAM), i.e. EAMSCR [0.5]
- It needs to be clarified to the employer that AAMSCR and EAMSCR will be more or less than the cost of accrual of the benefit for a year for which it is being charged (depending on whether $i > e$ or $i < e$) [0.5]
- Further the salary escalation assumption in calculation of PUMSCR or AAMSCR or EAMSCR may need to be used considering individual circumstances, if possible [0.5]
- Also the mortality and withdrawal rate assumption may be ignored [0.5]
- The PUMSCR/AAMSCR/EAMSCR needs to be calculated periodically, say after every 3 or 5 years in view of the changes in economic scenarios [0.5]

**[Max 10]
[50 Marks]**
